

上海交通大学答题纸

班级

课程名称

数字电子技术

我承诺，我将严格遵守考试纪律。

承诺人：

PART I

题号	1	2	3	4	5	6	7	8	9	10
得分	T	F	T	T	F	T	T	F	T	T
批阅人(流水阅卷教师签名处)			F			F			F	

Part II.

1. 1/10

PART II

1 2 3 4 5 6 7 8 9 10
C. D B C D D B C C. A2. ~~(211)~~ (1101 0001)₂

3. (1) 00101100

(2) 10^{-3} s /ms(3) 10^{-3} s 0.125 ms

4. 13

5. $\overline{A+B} = \overline{A} \cdot \overline{B}$

6. 200, 20kHz

7. 33.3

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(2021 至 2022 学年 第 2 学期)

课程名称 数电 E

PART IV

1. $Y = \sum m_i (i = 1, 4, 5, 7)$

$$= \bar{A}_3 \bar{A}_2 \bar{A}_1 A_0 + \bar{A}_3 A_2 \bar{A}_1 \bar{A}_0 + \bar{A}_3 A_2 \bar{A}_1 A_0 + \bar{A}_3 A_2 A_1 \bar{A}_0$$

$\begin{matrix} Y \\ A_3 A_2 \end{matrix}$	$\begin{matrix} A_1 A_0 \\ 00 \end{matrix}$	$\begin{matrix} 01 \end{matrix}$	$\begin{matrix} 11 \end{matrix}$	$\begin{matrix} 10 \end{matrix}$
00	0	1	0	0
01	1	1	1	0
11	x	x	x	x
10	0	0	x	x

min
SOP

$$Y = A_2 \bar{A}_1 + \bar{A}_3 \bar{A}_1 A_0 + A_2 A_0$$

$\begin{matrix} Y \\ A_3 A_2 \end{matrix}$	$\begin{matrix} A_1 A_0 \\ 00 \end{matrix}$	$\begin{matrix} 01 \end{matrix}$	$\begin{matrix} 11 \end{matrix}$	$\begin{matrix} 10 \end{matrix}$
00	0	1	0	0
01	1	1	1	0
11	x	x	x	x
10	0	0	x	x

min POS

$$Y = \bar{A}_2 \bar{A}_0 + A_3 \bar{A}_1 + \bar{A}_2 A_1 + A_1 \bar{A}_0$$

$$= (A_2 + A_0)(\bar{A}_3 + \bar{A}_1)(A_2 + \bar{A}_1)(\bar{A}_1 + A_0)$$

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Truth Table.

$A_1 A_0$	$B_1 B_0$	Y
00	00	0
00	01	0
00	10	0
00	11	0
01	00	1 ✓
01	01	0
01	10	0
01	11	0
10	00	1 ✓
10	01	1 ✓
10	10	0
10	11	0
11	10	1 ✓
11	11	0
11	00	1 ✓
11	01	1 ✓

$$Y = \bar{A}_1(A_0\bar{B}_1\bar{B}_0) + A_1(\bar{A}_0\bar{B}_1\bar{B}_0) + A_1(\bar{A}_0\bar{B}_1B_0) + A_1(A_0B_1\bar{B}_0) + A_1(A_0\bar{B}_1\bar{B}_0) + A_1(A_0\bar{B}_1B_0)$$

$$\bar{A}_1: (100)_2 = (4)_{10}$$

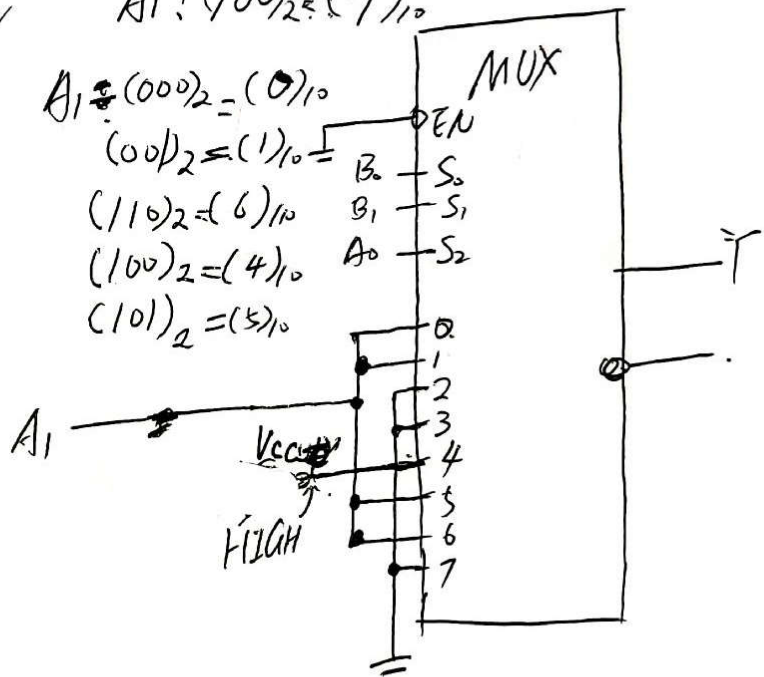
$$A_1: (000)_2 = (0)_{10}$$

$$(001)_2 = (1)_{10}$$

$$(110)_2 = (6)_{10}$$

$$(100)_2 = (4)_{10}$$

$$(101)_2 = (5)_{10}$$



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课程名称 E

$$Q^{n+1} = J \bar{Q}^n + K Q^n$$

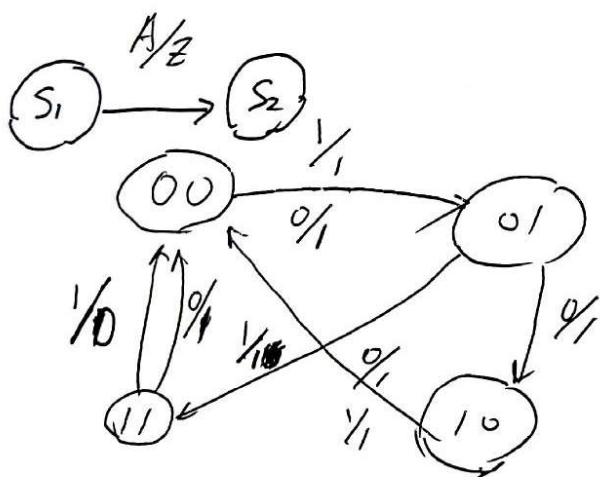
$$3. (1) J_0 = \bar{Q}_1^n, K_0 = \bar{A} \bar{Q}_1^n, Q_0^{n+1} = \bar{Q}_1^n \bar{Q}_0^n + A \bar{Q}_1^n Q_0^n$$

$$Q_0^{n+1} = \bar{Q}_1^n (\bar{Q}_0^n + A Q_0^n) = \bar{Q}_1^n (\bar{Q}_0^n + A)$$

$$(2) J_1 = Q_0^n, K_1 = 1$$

$$Q_1^{n+1} = Q_0^n \bar{Q}_1^n$$

$$(3) Z = \bar{A} Q_0^n Q_1^n$$

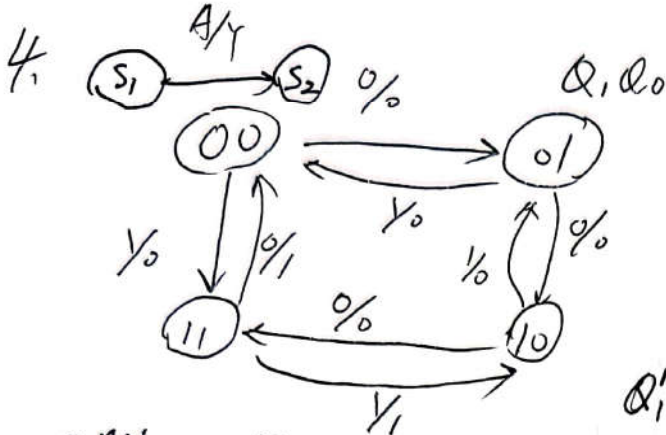


A	Q_1^n	Q_0^n	Q_1^{n+1}	Q_0^{n+1}	Z
0	0	0	0	1	1
0	0	1	1	0	1
0	1	0	0	0	1
0	1	1	0	0	1
1	0	0	0	0	1
1	0	1	1	0	1
1	1	0	0	0	0
1	1	1	0	0	0

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$$Q_1^{n+1} = J_1 \bar{Q}_1^n + \bar{K}_1 Q_1^n$$

$Q_1^n Q_0^n$	00	01	11	10
A	0	1	0	1
Y	1	0	1	0

$$Q_1^{n+1} = \bar{A} \bar{Q}_1^n Q_0^n + \bar{A} Q_1^n \bar{Q}_0^n + A \bar{Q}_1^n \bar{Q}_0^n + A Q_1^n Q_0^n$$

$$Q_1^{n+1} = (\bar{A} Q_0^n + A \bar{Q}_0^n) \bar{Q}_1^n + (\bar{A} \bar{Q}_0^n + A Q_0^n) Q_1^n$$

$$J_1 = A \oplus Q_0^n = K_1 = A \oplus Q_0^n$$

$Q_1^n Q_0^n$	00	01	11	10
A	0	1	0	1
Y	1	0	1	0

$$Q_0^{n+1} = \bar{Q}_0^n \quad J_0 = 1 \quad K_0 = 1$$

$Q_1^n Q_0^n$	00	01	11	10
A	0	0	1	0
Y	1	0	1	0

$$Y = Q_1^n Q_0^n$$

